

Finite-Blocklength Latency Minimization for Uplink and Downlink Communications: Method Results: Interference

July 6, 2026

Abstract

This document collects the exported interference figures for uplink and downlink under both payload completion and fixed block targets. Throughout, the *Training Only Method* is used as the benchmark for the *Training + Testing Method*. The uplink figures mostly show moderate interference reshaping inside a similar operating regime. The downlink figures are more structural, especially when users are removed from the active set and their traces drop toward -300 dB.

1 Scope and Naming

This document is intentionally interference-only. The terminology is aligned with the companion theory report:

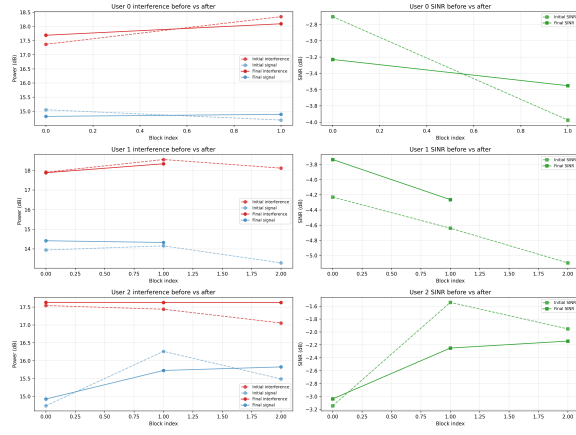
- **Training Only Method:** direct online optimization during scheduling,
- **Training + Testing Method:** a separate training stage followed by testing-stage inference,
- **Payload completion:** the scenario where each user continues until its remaining payload is exhausted,
- **Fixed block targets:** the scenario with a prescribed blockwise service pattern over a fixed horizon.

2 Uplink Payload Completion

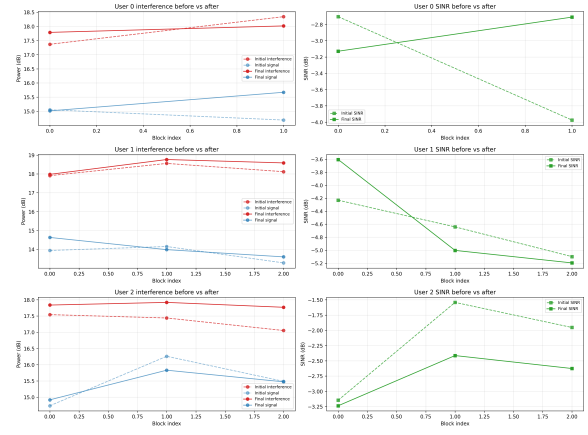
The uplink payload-completion interference picture stays fairly close across the two methods. The benchmark does improve final SINR for users 1 and 2, but the overall difference remains moderate rather than dramatic.

Table 1: Uplink payload-completion per-user SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	$-3.359/ - 3.396$ dB	$-3.359/ - 2.908$ dB	Training + Testing is better on this user, but not enough to change the system outcome.
1	$-4.649/ - 4.006$ dB	$-4.649/ - 4.592$ dB	Benchmark has the clearest SINR gain here.
2	$-2.173/ - 2.459$ dB	$-2.173/ - 2.743$ dB	Benchmark ends less negative.

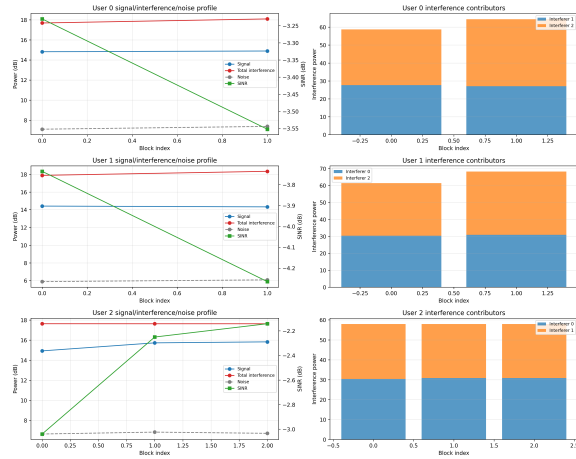


(a) Training Only Method

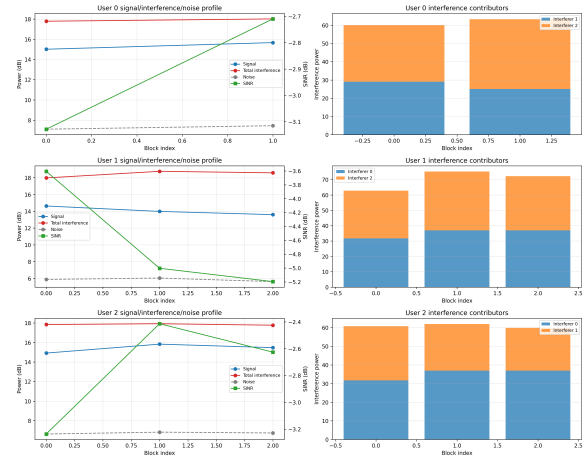


(b) Training + Testing Method

Figure 1: Uplink payload-completion per-user interference before/after comparison.

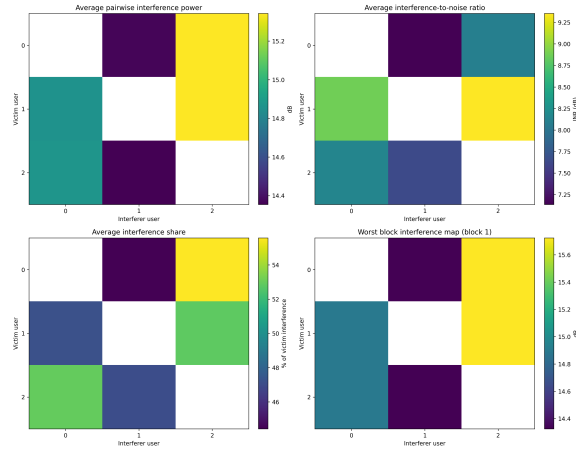


(a) Training Only Method

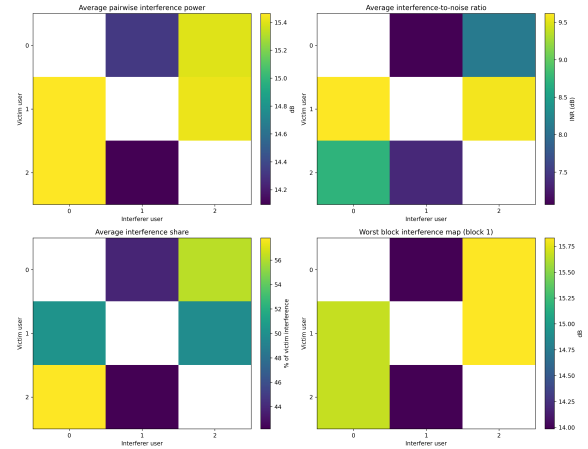


(b) Training + Testing Method

Figure 2: Uplink payload-completion per-user interference profiles.



(a) Training Only Method



(b) Training + Testing Method

Figure 3: Uplink payload-completion interference heatmaps.

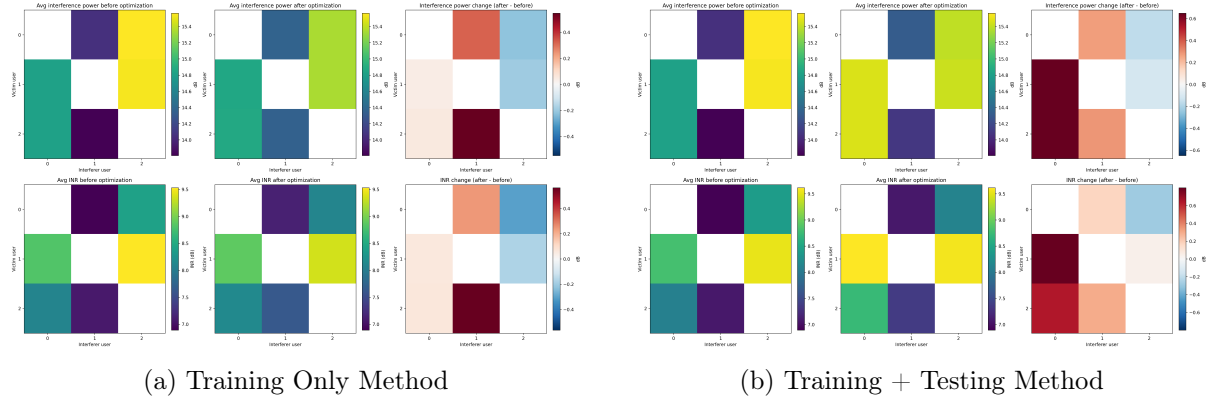


Figure 4: Uplink payload-completion before/after heatmap comparison.

3 Uplink Fixed Block Targets

The uplink fixed-target case remains close across methods as well. All users are fully served, and the interference differences mainly appear as small SINR redistributions.

Table 2: Uplink fixed-block-target per-user SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	-2.647/ - 3.093 dB	-2.647/ - 2.807 dB	Training + Testing is slightly better.
1	-4.329/ - 3.981 dB	-4.329/ - 4.175 dB	Benchmark is slightly better.
2	-3.071/ - 2.896 dB	-3.071/ - 3.039 dB	Benchmark is slightly better.

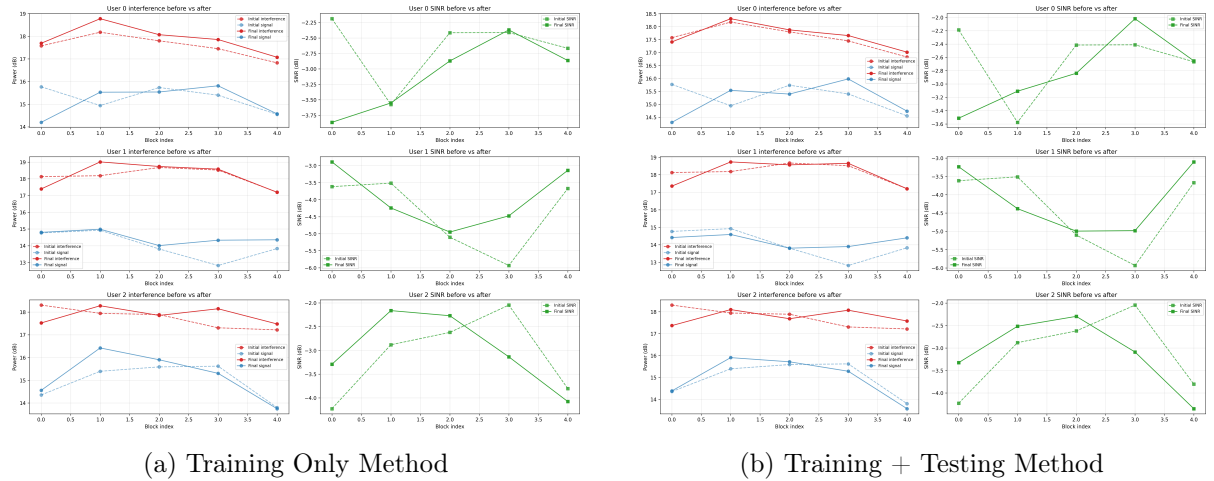
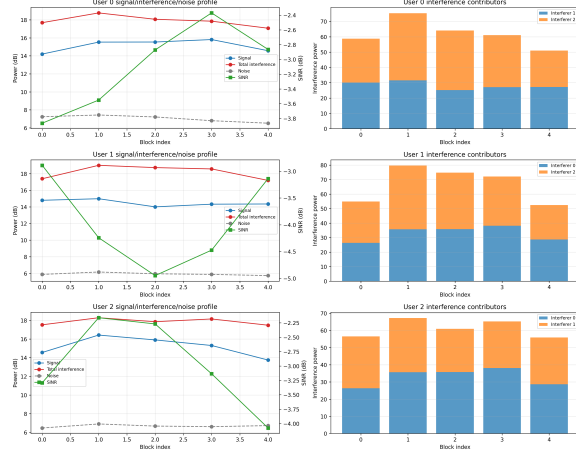
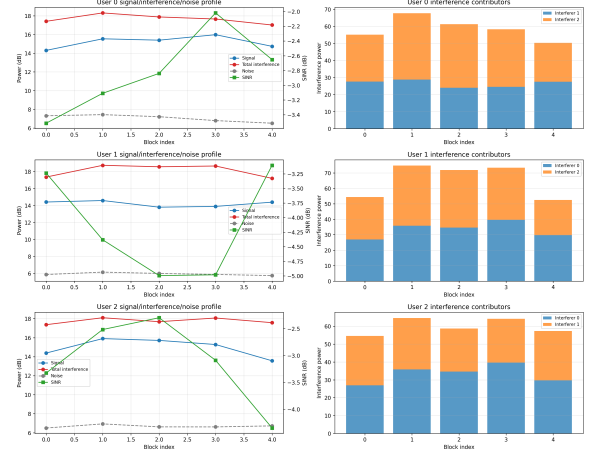


Figure 5: Uplink fixed-block-target per-user interference before/after comparison.

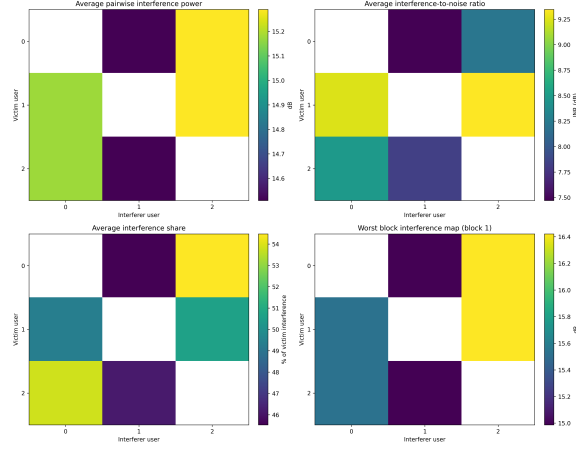


(a) Training Only Method

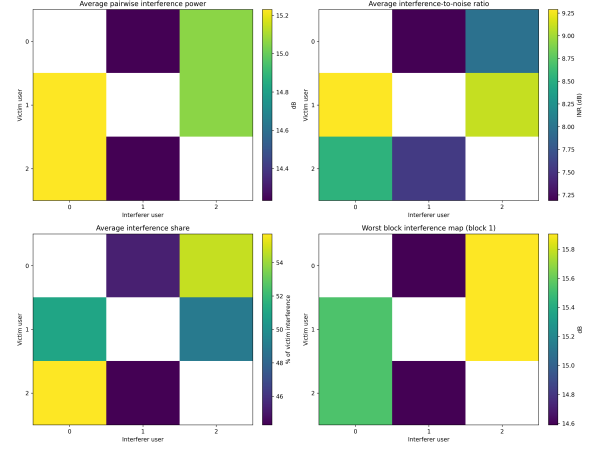


(b) Training + Testing Method

Figure 6: Uplink fixed-block-target per-user interference profiles.

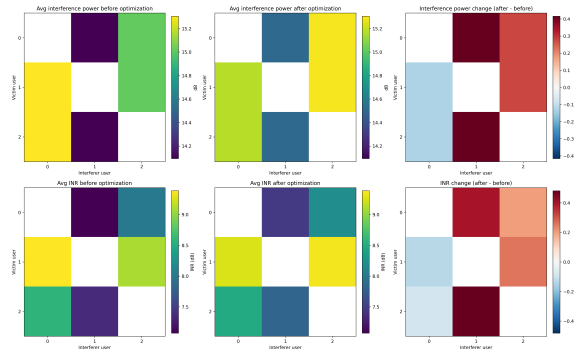


(a) Training Only Method

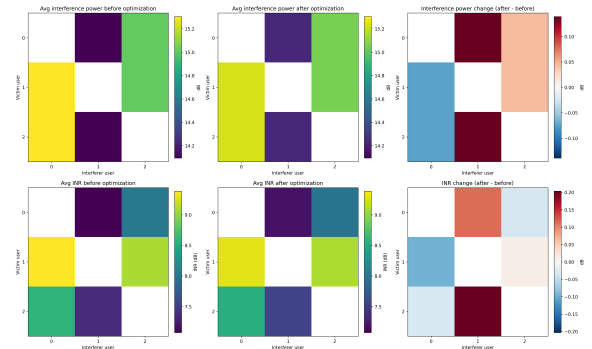


(b) Training + Testing Method

Figure 7: Uplink fixed-block-target interference heatmaps.



(a) Training Only Method



(b) Training + Testing Method

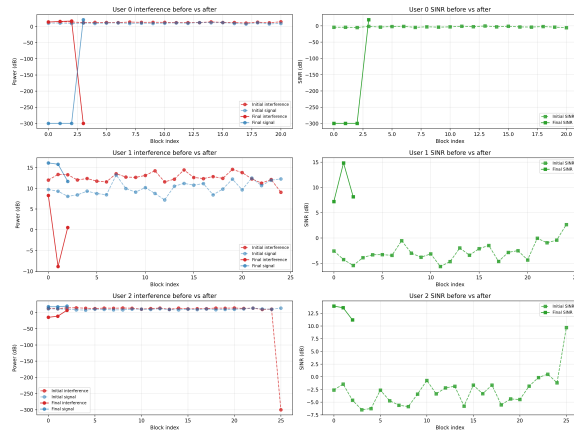
Figure 8: Uplink fixed-block-target before/after heatmap comparison.

4 Downlink Payload Completion

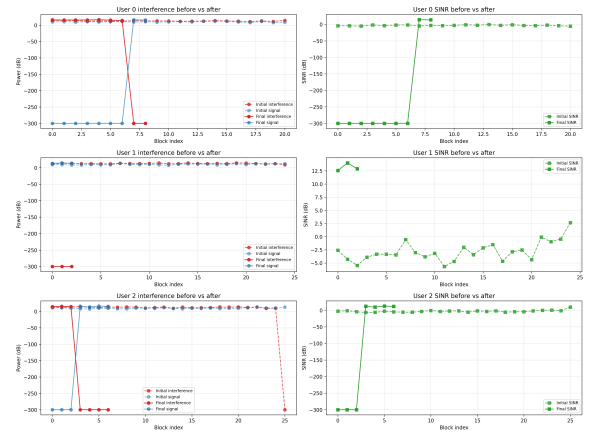
The downlink payload-completion interference results are more structural. The benchmark clears users from the active set more decisively, and that shows up in both the -300 dB inactive segments and the shorter schedule tail.

Table 3: Downlink payload-completion per-user served-block SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	$-3.003/18.920$ dB	$-3.003/14.017$ dB	Benchmark is much stronger after skipped-block concentration.
1	$-2.774/10.052$ dB	$-2.774/13.141$ dB	Training + Testing is stronger on this easier user.
2	$-2.764/12.894$ dB	$-2.764/11.416$ dB	Benchmark is stronger and finishes earlier.

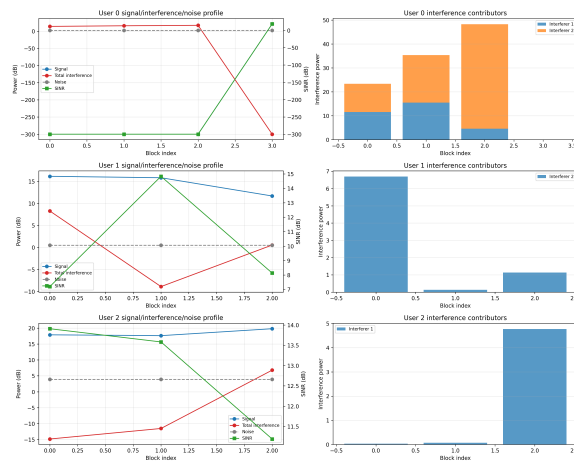


(a) Training Only Method

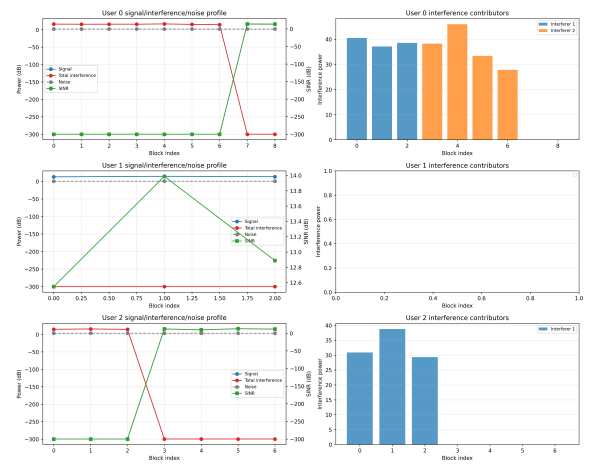


(b) Training + Testing Method

Figure 9: Downlink payload-completion per-user interference before/after comparison.



(a) Training Only Method



(b) Training + Testing Method

Figure 10: Downlink payload-completion per-user interference profiles.

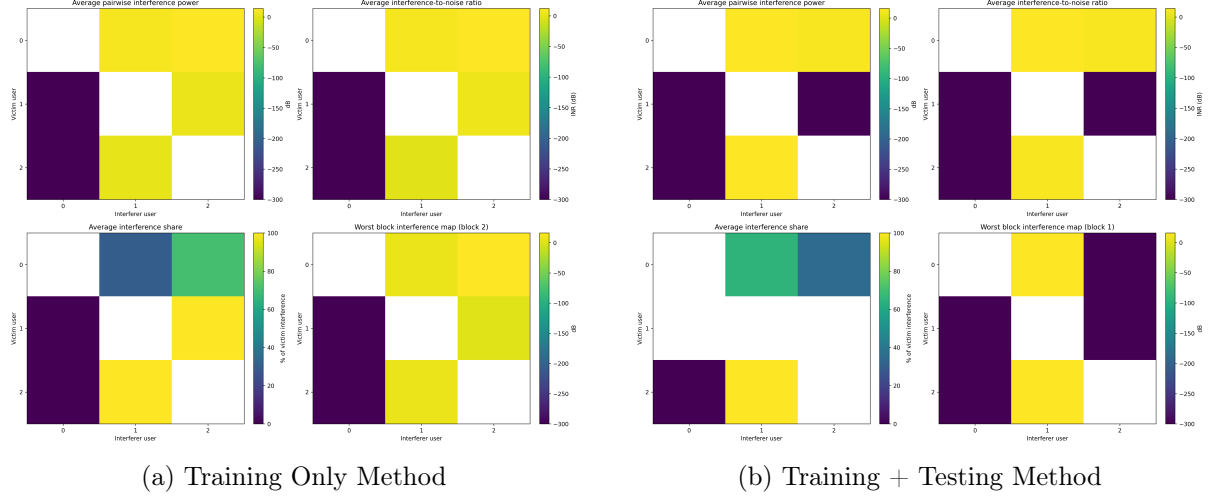


Figure 11: Downlink payload-completion interference heatmaps.

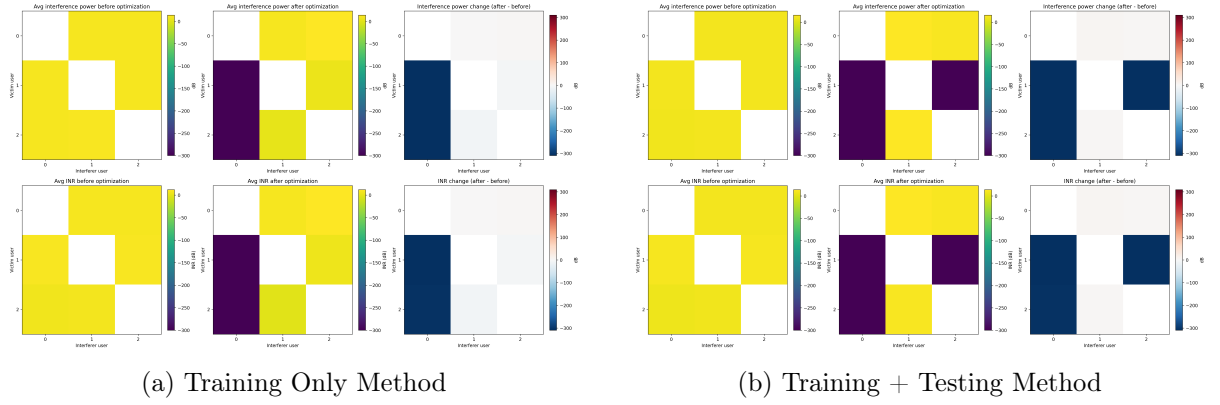


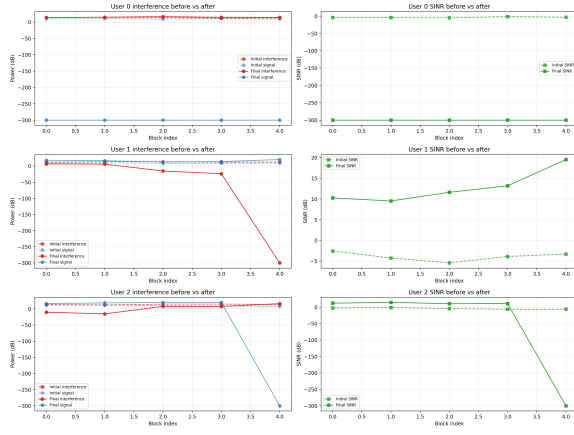
Figure 12: Downlink payload-completion before/after heatmap comparison.

5 Downlink Fixed Block Targets

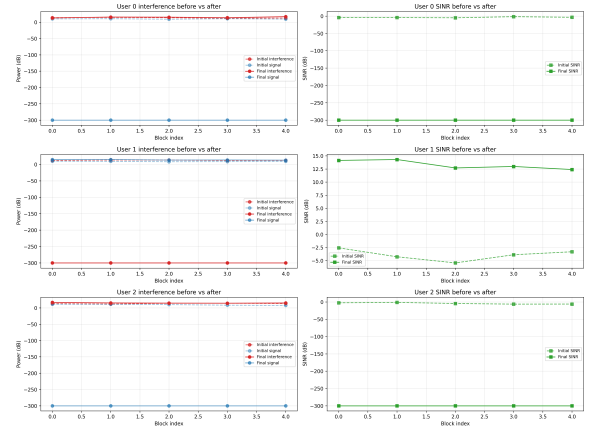
The downlink fixed-target case is the most feasibility-limited interference setting. Users 0 and 2 can collapse to -300 dB final block SINR when they receive no effective service, while user 1 remains the only consistently recoverable user.

Table 4: Downlink fixed-block-target per-user served-block SINR summary

User	Training Only init/final	Training + Testing init/final	Reading
0	$-3.770/-300.000$ dB	$-3.770/-300.000$ dB	No service under either method.
1	$-3.904/12.801$ dB	$-3.904/13.300$ dB	Both recover this user well; Training + Testing is slightly higher.
2	$-4.298/11.894$ dB	$-4.298/-300.000$ dB	Benchmark still serves this user partially, while Training + Testing fails completely.

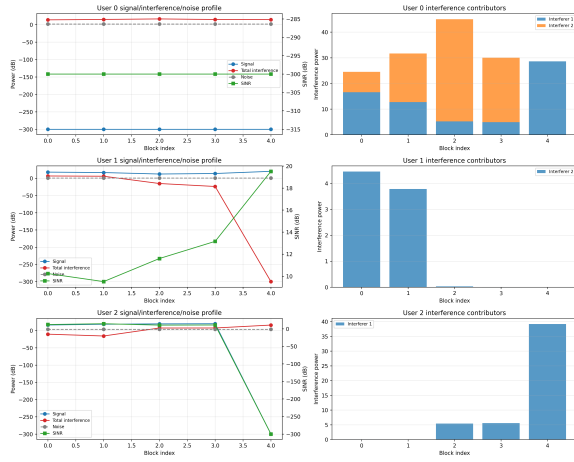


(a) Training Only Method

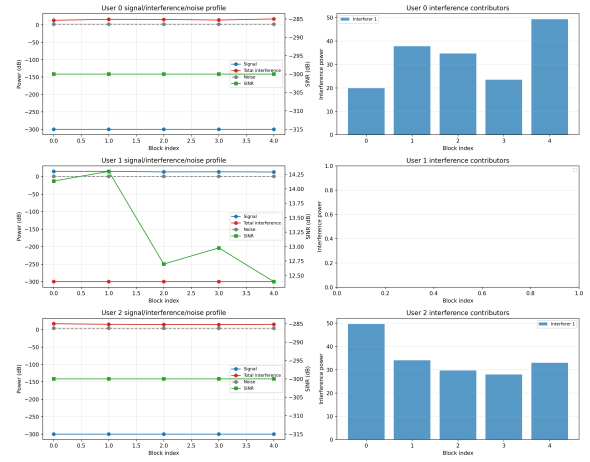


(b) Training + Testing Method

Figure 13: Downlink fixed-block-target per-user interference before/after comparison.

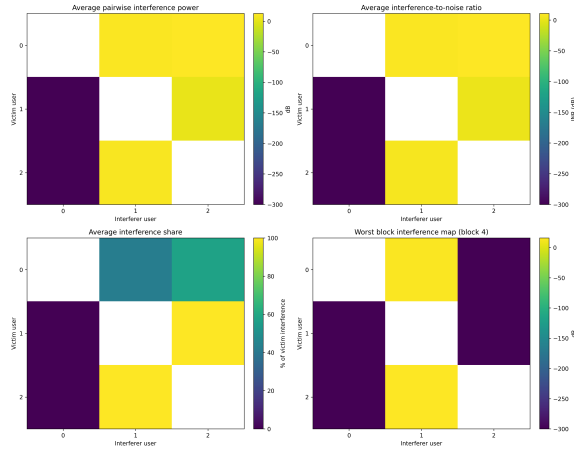


(a) Training Only Method

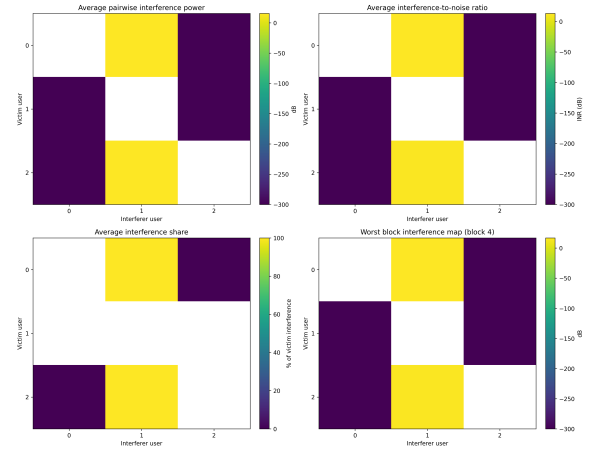


(b) Training + Testing Method

Figure 14: Downlink fixed-block-target per-user interference profiles.



(a) Training Only Method



(b) Training + Testing Method

Figure 15: Downlink fixed-block-target interference heatmaps.

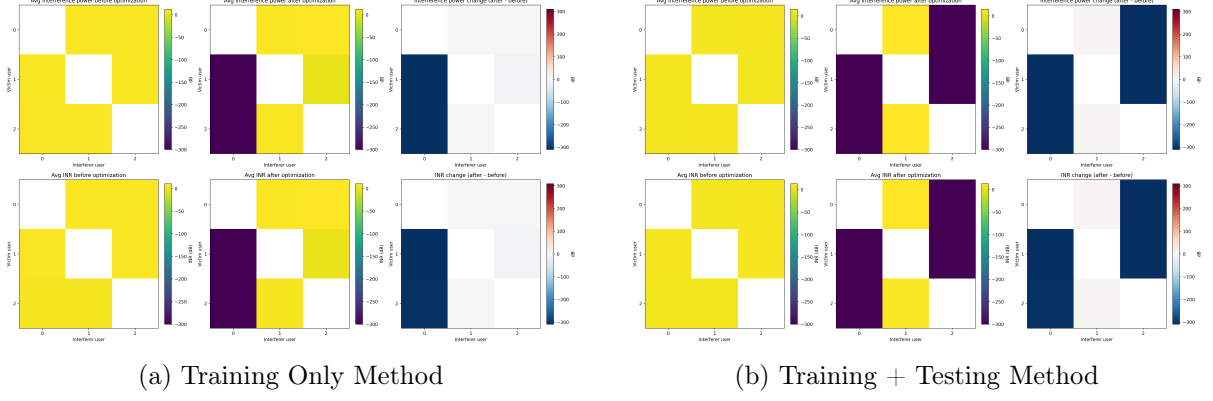


Figure 16: Downlink fixed-block-target before/after heatmap comparison.

6 Interference Takeaways

Across all exported figures, the uplink remains the closer match between methods. Its interference figures mostly show small SINR redistributions inside a similar operating range. The downlink is more revealing. In payload completion, the benchmark concentrates service more decisively on the currently useful users. In fixed block targets, the same coordination problem becomes a feasibility issue, most clearly on downlink user 2.